Name ____________________

6 Topic Assessment Form B

1. Which function g represents the exponential function $f(x) = 3^x$ after a horizontal stretch by a factor of 4 and a reflection across the y-axis?

Ⓐ $g(x) = (3)^{-\frac{x}{4}}$

Ⓑ $g(x) = (3)^{-4x}$

Ⓒ $g(x) = -(3)^{\frac{x}{4}}$

Ⓓ $g(x) = -(3)^{4x}$

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2. A population of mice increases from 400 to 1400 over 7 years. Which function best models the population x years from the first measurement?

Ⓐ $f(x) = 400\left(1 + 2.5^{\frac{1}{7}}\right)^x$

Ⓑ $f(x) = 400(1 + 2.5)^{\frac{1}{7}x}$

Ⓒ $f(x) = 400\left(1 + \frac{2.5}{7}\right)^x$

Ⓓ $f(x) = 400\left(1 + \frac{2.5}{7}\right)^{7x}$

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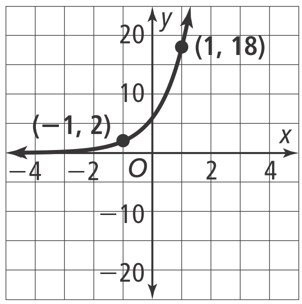
3. In the expression shown $t \neq s$.

$$\frac{(s - t) + (2s - 2t)^2}{t - s}$$

When x replaces $(s - t)$ which expression is equivalent to the expression shown. Select all that apply.

- [x] **A.** $\frac{x + 4x^2}{-x}$
- [ ] **B.** $\frac{x - 2x^2}{x}$
- [ ] **C.** $\frac{x + (4x)^2}{x}$
- [x] **D.** $\frac{-x - 4x^2}{x}$
- [ ] **E.** $\frac{x + (2x)^2}{x}$

4. Which of the following functions has a slower growth rate than the function shown in the graph?



Ⓐ $f(x) = 4(4.5)^x$

Ⓑ $f(x) = 6(3.5)^x$

Ⓒ $f(x) = 8(3.1)^x$

Ⓓ $f(x) = 10(2)^x$

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5. The height of a launched weather balloon can be represented by an exponential model, $y = a \cdot b^x$, where y is the height in km after x min. After 2 min, the height is 3 km, and after 3 min it is 9 km. Which represents another point on the graph of this model?

Ⓐ $(1, -3)$ Ⓒ $(4, 15)$

Ⓑ $(1, 1)$ Ⓓ $(4, 16)$

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6. A school of fish has a population of 200. The population is decreasing at a rate of 5% per year. Select the model of the population after t years that reveals the quarterly decay factor.

Ⓐ $y = 200(0.95)^t$

Ⓑ $y = 200(0.9875)^{4t}$

Ⓒ $y = 200(0.9875)^t$

Ⓓ $y = 200(0.9873)^{4t}$

7. What function is the inverse of the exponential function $y = \left(\frac{1}{3}\right)^x$?

Ⓐ $y = x^{\frac{1}{3}}$ Ⓒ $y = \log_{\frac{1}{3}} x$

Ⓑ $y = \left(\frac{1}{3}\right)^x$ Ⓓ $y = \log_x \left(\frac{1}{3}\right)$

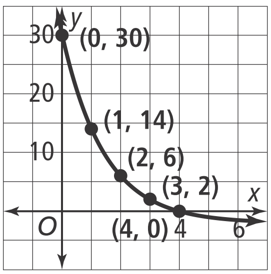
8. What is the value of n in the formula $S = \frac{a_1(1 - r^n)}{(1 - r)}$ for the sum of the geometric series $0.4 + 1.2 + 3.6 + \ldots + 7873.2$?

$n =$ 10

9. What is the solution to the equation $\log_3 (2x - 3) = -1$? Round to the nearest thousandth.

$x =$ 1.667

10. The graph shows the function $f(x) = 32\left(\frac{1}{2}\right)^x - 2$. What is the value of the inverse function f^{-1} at $x = 2$?



$f^{-1}(2) =$ 3

11. What is the asymptote of the graph of $g(x) = \log_2 (x + 4)$?

Ⓐ vertical at $x = 0$

Ⓑ vertical at $x = -4$

Ⓒ horizontal at $y = 0$

Ⓓ horizontal at $y = -4$

12. Complete the equation of the inverse of the function $f(x) = \log_4 (3x)$.

| $\frac{1}{3}$ | 3 | 4 | $\frac{1}{4}$ |

$f^{-1}(x) = \frac{1}{3}(4)^x$

13. Use the properties of logarithms to choose the expression $x\log 2 - \log \frac{1}{3}$ as a single logarithm.

Ⓐ $\log\left(2x - \frac{1}{3}\right)$ Ⓒ $\log\left(2^x - \frac{1}{3}\right)$

Ⓑ $\log\left(\frac{2}{3x}\right)$ Ⓓ $\log(3(2)^x)$

14. The profit for a new website, in thousands of dollars, can be represented by 5^x, where x is time in months. Find the time when the company's profit is \$15,000. Select the exact solution as a logarithm, and select an approximate solution.

- [ ] A. $x = \log_{15}5$
- [x] D. $x = 1.689$
- [x] B. $x = \log_5 15$
- [ ] E. $x = 0.594$
- [ ] C. $x = \log 15$
- [ ] F. $x = 0.176$

15. What is the sum of the geometric series $S = 1.375 + 1.1 + 0.88 + \ldots + 0.031$? Round to the nearest thousandth.

$S =$ 6.751

16. Complete the steps to write the expression $2\log a + 2\log b - 3\log c$ as a single logarithm using the properties of logarithms.

| Product | Quotient | Power |

$2\log a + 2\log b - 3\log c$

$= \log a^2 + \log b^2 - \log c^3$ Power

$= \log a^2b^2 - \log c^3$ Product

$= \log\left(\frac{a^2b^2}{c^3}\right)$ Quotient

17. For the function $f(x) = 2 \cdot (3)^{-x}$, identify the y-intercept and asymptote.

y-Intercept: 2

Asymptote: $y =$ 0